

Question 1: What is the actual name of the address that programmers see on an x86 processor? What specific information does this address contain that is necessary for address translation? **[2 marks]**

Answer:

Logical address [+1 marks]

Segment selector [+1 marks]

Question 2: Paging may result in internal fragmentation. True/False. **Justify** using an example. **[2 marks]**

Answer:

True. See Lecture 17, slide 6.

No partial marks. Marks only if correct example (may not be same as in slide).

Question 3: Which of the two segment registers on x86 processor hold the **most** important information related to the program execution? **Justify**, and provide their full name. **[2 marks]**.

Answer:

Code segment (CS) as it contains information related to .text section. [+1 marks]

Stack segment (SS) as it contains information related to the execution stack of the process. [+1 marks]

Question 4: A machine has a 16-bit address space that programmers can use. Each CPU has a dedicated cache of 4KB to store the page table of a running process. The size of an integer or pointer on this machine is 4 bytes. To maximize the number of entries in the page table stored in the cache, how many bits in a 16-bit address should be used to identify the pages uniquely? **Justify**. **[2 marks]**

Answer:

Maximum size of page table to ensure cache can entirely store the page table = 4KB. [+0.75 marks]

Assuming an array-based implementation of page table, each entry will be of 4b. Hence, total number of page table entries that could fit in the cache = $4\text{KB} / 4 = 2^{10}$. [+0.75 marks]

Total bits required to uniquely identify these pages = 10. [+0.5 marks]

Question 5: Calculate the total number of page faults that must happen while accessing the heap space as shown in the program. Assume size of integer as 4 bytes and the page size to be 64 bytes. **Justify**. **[2 marks]**.

Answer:

Page fault will happen on the pages being accessed. [+1 marks]

Hence, total page faults for accessing 128 integers = $(128 * 4) / 64 = 8$

[+1 marks]

Page table size = 2^{12}

$4N = 2^{12}$

$N = 2^{10}$

```
int main( ) {
    int* a = (int*) malloc(1024);
    for(int i=0; i<128; i++)
        a[i] = 0;
    return 0;
}
```